

Ran Chen

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Education

University of Illinois Urbana-Champaign

May 2028

Bachelor of Science in Aerospace Engineering

GPA: 4.00/4.00

Shanghai World Foreign Language Academy

May 2024

International Baccalaureate Diploma

GPA:41/45

Experience

Institute for Nano- and Microfluidics

Technische Universität Darmstadt, Germany

Research Assistant

May 2026 – Present

- Designed a novel fiber-laden T-junction experiment using a microfluidic plate formulated by PMMA and solvent bonding
- Established analytical physical model to describe the bridging phenomenon observed and developed image analysis workflow to cross-validate the model

Computer-Aided Design Course (ME170)

University of Illinois, Urbana-Champaign, USA

Course Assistant

September 2025 – Present

- Performed weekly teaching duty and helped students with mastering CAD software Fusion 360 and adhering with ASME and ISO standards
- Developed novel conceptual questions and rebuilt the course on the online platform PrairieLearn

Complex Fluids and Turbulence Laboratory

University of Illinois, Urbana-Champaign, USA

Research Assistant

January 2025 – Present

- Designed a 3D Convolutional Neural Network with Dense Block to perform feature detection to transition plane Poiseuille flow
- Designed methodology to determine the Exact Coherent States (ECS) of transition plane Poiseuille flow by performing Proper Orthogonal Decomposition (POD) and utilizing Implicit Rank-Minimizing Autoencoder with Neural ODE

Design, Build, Fly

University of Illinois, Urbana-Champaign, USA

Team member

September 2024 – September 2025

- Innovated, designed and improved novel parts for the plane used for competition with a focus on light weight design and the most optimum aerodynamic effect
- Actively participated in CAD design meeting and provided feedback for team evaluation

Honors, Awards and Test Scores

Scholarship of Excellence: 3,099 EUR (2026)

Illinois for Illinois Scholarship: 2,000 USD (2026)

Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

Tan Family Education Foundation Scholarship: 10,000 USD (2025), 3,000 USD (2026)

Top Gold, British Physics Olympic (2023)

SAT: 1510/1600 (Reading and Writing:710/800; Mathematics:800/800)

TOEFL: 112/120 (Reading: 29/30; Listening: 30/30; Speaking 24/30; Writing: 29/30)

Skills

Fusion 360; HPC; Latex; Linux; MATLAB; Python; Siemens NX; PyTorch; HTML